

ZEWAIL CITY OF SCIENCE AND TECHNOLOGY

Faculty of Engineering

Project Report

Simulation of Rapid Thermal Processing (RTP)

Using COMSOL Multiphysics

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1. Introduction

Rapid Thermal Processing is an important technique in making semiconductors. It is different from the way of using a furnace to heat things up. This old way makes the wafers get very hot for a time. Rapid Thermal Processing uses infrared light to heat up a semiconductor wafer very quickly. Usually in just a few seconds. It gets very hot over 1000 degrees Celsius. Then it cools down just as fast. This way of doing things helps keep the stuff inside the wafer from moving around too much. It also helps keep the wafer from getting too hot for long which makes the devices work better and more of them work properly.

This report looks at why Rapid Thermal Processing's so important for making semiconductors. It talks about how it works, how to design the systems what problems engineers might have and what parts are used to make it work. It also uses a computer program called COMSOL Multiphysics to simulate what happens in a Rapid Thermal Processing chamber. This simulation is, like a model that looks at how the infrared light heats up the wafer and how the temperature stays steady

2. Research on RTP Importance, Concepts, Design Approaches, and Key Challenges

2.1 Importance of RTP in Semiconductor Manufacturing

As semiconductor devices get smaller and smaller we need to be more precise when we work with heat. This is especially true when we are working with devices that're less than 5 nm in size. Rapid Thermal Processing or RTP helps with important things when we are making very small devices:

- **Dopant Activation:** When we put special chemicals into a device we need to make sure they are working properly. We do this by heating the device up quickly. RTP can do this in a few seconds, which helps keep the chemicals from moving around too much. This is important for making sure the devices work correctly.
- **Gate Dielectric Growth:** We need to be very careful when we are growing layers on our devices. These layers are very thin. Need to be grown at just the right temperature. RTP helps us do this without making a mess.
- **Silicide Formation:** Sometimes we need to combine metal and silicon to make a material. We can do this by heating it up quickly in an RTP system.
- **Thermal Budget Reduction:** When we are working with heat we need to be careful not to damage our devices. RTP helps us do this by using heat for a short time. This is better than using heat for a time, which can hurt our devices.
- **Single-Wafer Processing:** RTP works with one device at a time which makes it easier to control what is happening. We can also switch between recipes quickly which helps us make a lot of different devices.

Overall RTP is very important when we are making devices like computer chips and memory devices. We need to be able to control the heat carefully and RTP helps us do this. Semiconductor devices, like these need RTP to work properly. It is a very useful tool.

2.2 Fundamental Concepts of RTP

2.2.1 Heating Mechanism

The primary energy transfer mechanism in RTP is thermal radiation. High-intensity tungsten-halogen or arc lamps emit intense IR radiation in the wavelength range of 0.5 – 5 μm . Silicon wafers absorb this radiation efficiently, especially at temperatures above $\sim 600^\circ\text{C}$ where free-carrier absorption becomes significant.

Stefan-Boltzmann law governs radiative heat transfer:

$$q = \varepsilon \sigma T^4$$

where q is the radiated heat flux (W/m^2), ε is the emissivity of the surface,

$\sigma = 5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$ is the Stefan-Boltzmann constant, and T is the absolute temperature (K). At temperatures above 1000°C , radiation dominates all other heat transfer modes, contributing to approximately 90% of total heat transfer.

2.2.2 Rapid Heating and Cooling

Typical RTP ramp rates range from 10 to $250^\circ\text{C}/\text{s}$. Such rapid heating is achieved by the high power density of the lamp arrays (often exceeding 40 kW per system) and the low thermal mass of the thin silicon wafer (standard thickness: 0.5 – 0.775 mm for 200 – 300 mm wafers).

Cooling is equally critical. After processing, lamps are extinguished and the wafer cools by radiation (primarily) and convection. Cooling rates of $50 - 150^\circ\text{C}/\text{s}$ can be achieved. Rapid cooling prevents unwanted secondary reactions and "freezes" the desired material microstructure.

2.2.3 Design Approaches

Several design philosophies have been developed for RTP systems:

- Single-Zone Lamp Arrays: Simple design with uniform lamp density, suitable for processes with relaxed uniformity requirements.
- Multi-Zone Lamp Arrays: Independent control of lamp zones (center, edge, intermediate) enables dynamic compensation for temperature non-uniformity and edge cooling losses.
- Rotating Susceptor Design: The wafer or susceptor rotates during heating to average out lamp-induced non-uniformities and improve temperature uniformity to $\pm 1-2^\circ\text{C}$ across the wafer.

- **Reflective Chamber Geometry:** The chamber walls are coated with gold or polished aluminum to maximize reflectivity and uniformly distribute IR radiation onto the wafer.
- **Pyrometric Control:** In-situ optical pyrometers measure the wafer temperature at multiple points in real time, enabling closed-loop temperature control.

2.3 Key Challenges in RTP

Despite its advantages, RTP presents several engineering challenges that must be carefully managed:

2.3.1 Temperature Uniformity

Achieving within-wafer temperature uniformity (WIWU) is arguably the most critical challenge in RTP. Edge cooling — caused by thermal radiation losses at the wafer periphery — creates center-to-edge temperature gradients. These gradients cause non-uniform dopant activation, stress-induced slip lines (crystal defects), and device parameter variation. Solutions include multi-zone lamp control, rotating wafer designs, and edge ring guard rings that reduce the thermal gradient.

2.3.2 Wafer Slip and Thermal Stress

Rapid temperature ramps generate steep temperature gradients across the wafer thickness and radius, inducing thermal stress. When this stress exceeds the yield strength of silicon (which decreases above $\sim 800^{\circ}\text{C}$), slip dislocations form along crystallographic planes. Slip can cause wafer warpage, particle generation, and device failure. Careful ramp rate optimization and multi-zone control are used to mitigate slip.

2.3.3 Emissivity-Dependent Measurement Errors

Temperature measurement via pyrometry relies on accurate knowledge of the wafer's emissivity. However, silicon's emissivity varies significantly with temperature, doping concentration, film stack, and surface condition. Emissivity variations cause systematic errors in pyrometric temperature measurement ($\pm 20\text{--}50^{\circ}\text{C}$ in some cases), which can lead to process excursions. Dual-wavelength and emissivity-corrected pyrometers are employed to address this challenge.

2.3.4 Process Scalability and Integration

As wafer diameters increase from 200 mm to 300 mm (and future 450 mm), maintaining temperature uniformity across a larger area becomes proportionally more difficult. Additionally, integrating RTP into clustered multi-chamber systems (e.g., for in-situ pre-clean + anneal sequences) requires careful interface design to prevent wafer contamination and maintain processing speed.

2.3.5 Lamp Aging and Maintenance

Tungsten-halogen lamps degrade over time due to tungsten evaporation and filament thinning. This aging changes the lamp's spectral output and power delivery, potentially causing process drift. Lamp monitoring systems and periodic calibration are required to ensure process reproducibility.

2.4 RTP System Components and Material Choices

An RTP system consists of several critical components. The selection of materials for each is driven by thermal, optical, chemical, and mechanical requirements:

Table 1: RTP System Components and Material Selection Rationale

Component	Material	Reason for Selection
IR Lamp Array	Tungsten-Halogen / Arc Lamps	Tungsten filaments emit intense broadband radiation peaking at $\sim 1\ \mu\text{m}$, efficiently absorbed by silicon. Halogen cycle regenerates the filament, extending lamp life. High power density (up to $100\ \text{W}/\text{cm}^2$) enables rapid wafer heating.
Process Chamber	Stainless Steel / Aluminum	High structural strength and machinability. Stainless steel provides corrosion resistance under process gas environments. Internal surfaces are polished or gold-coated to maximize IR reflectivity and distribute heat uniformly.
Quartz Window / Tube	Fused Quartz (SiO_2)	Fused quartz transmits IR radiation ($0.2 - 3.5\ \mu\text{m}$ wavelength) with $>90\%$ transmittance, allowing lamp energy to reach the wafer. It can withstand temperatures up to 1100°C continuously and resists thermal shock. Chemically inert under most process gases.
Wafer Susceptor / Support	SiC-Coated Graphite	SiC coating provides a chemically inert, high-emissivity surface ($\epsilon \approx 0.9$) for

		uniform heat distribution. Graphite core offers high thermal conductivity and low thermal mass. SiC prevents carbon contamination of the wafer.
Chamber Wall Coating	Gold or Polished Al	Gold provides >98% reflectivity in the near-IR, maximizing utilization of lamp energy and improving temperature uniformity across the wafer surface.
Temperature Sensor	Optical Pyrometer (Si or InGaAs)	Non-contact measurement avoids wafer contamination. Measures thermal emission from the wafer surface at specific wavelengths (e.g., 0.95 μm or 1.55 μm). Enables real-time closed-loop temperature control.
Silicon Wafer	Single-Crystal Silicon (p or n type)	The device substrate. Absorbs IR radiation efficiently. Thin wafer (~0.5 mm) provides low thermal mass enabling rapid temperature ramp rates. Standard resistivity and orientation selected per process requirements.
Process Gas Lines	Electropolished 316L SS	Prevents contamination and corrosion under oxidizing or reducing atmospheres (O_2 , N_2 , H_2 , Ar, etc.). Electropolishing minimizes surface roughness and particle generation.

3. COMSOL Multiphysics Simulation for Thermal Analysis

3.1 Simulation Overview and Objectives

The simulation models a simplified 2D cross-section of an RTP chamber using COMSOL Multiphysics. The primary objectives are:

- To visualize the transient and near-steady-state temperature distribution within the RTP chamber, including the silicon wafer, quartz tube, and surrounding chamber, at $t = 60$ s.
- To quantify the heating efficiency of the IR lamp array and its effect on wafer temperature.
- To analyze surface-to-surface radiation as the dominant heat transfer mechanism at elevated temperatures.
- To plot the temperature at the midpoint of the silicon wafer as a function of time (0–60 s) under time-dependent conditions.

3.2 Model Geometry and Dimensions

The 2D cross-sectional model represents a circular RTP chamber with concentric features. The geometry was constructed in COMSOL according to the following dimensions specified in the project requirements:

Table 2: Model Geometry Dimensions

Component	Dimension
RTP Chamber Radius	120 mm
IR Lamp Radius	10 mm
Quartz Tube Outer Radius	60 mm
Quartz Tube Inner Radius	55 mm
Silicon Wafer Width	50 mm
Silicon Wafer Thickness	0.5 mm

The IR lamps are arranged in a circular pattern at a radial distance of 80 mm from the chamber center, positioned between the quartz tube outer wall and the RTP chamber wall, as illustrated in Figure 1.

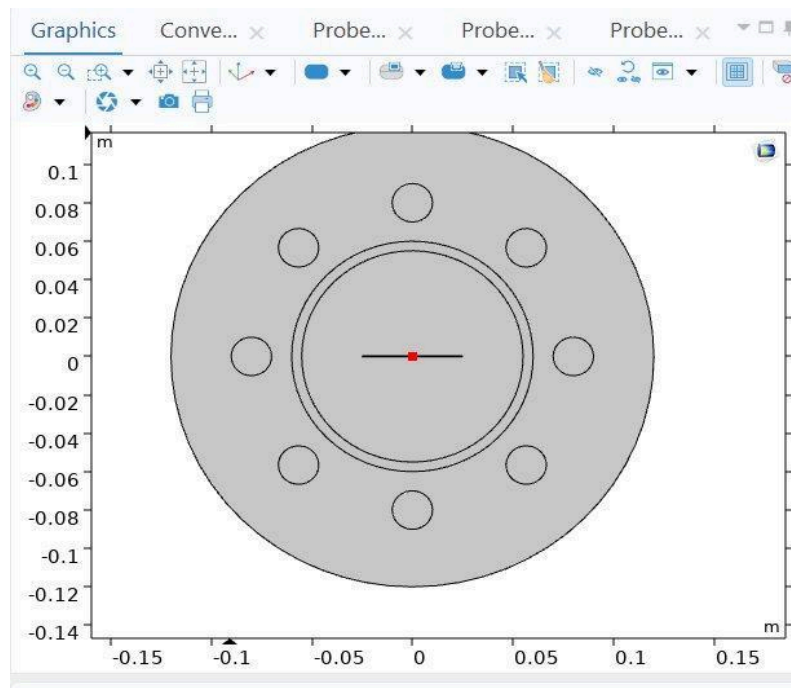


Figure 1: COMSOL model geometry — 2D cross-section of the RTP chamber showing the silicon wafer, quartz tube, and IR lamp arrangement.

3.3 Boundary Conditions and Physics Setup

3.3.1 Heat Transfer Physics

The simulation utilizes the Heat Transfer in Solids module in COMSOL, coupled with Surface-to-Surface Radiation. Since RTP operates at temperatures exceeding 1000°C, radiative heat transfer dominates, accounting for approximately 90% of total heat transfer as per the Stefan-Boltzmann law. Conductive heat transfer is also modeled within solid domains.

3.3.2 Boundary Conditions

The following boundary conditions were applied to the model:

- Outer Chamber Wall (RTP Chamber Outer Radius = 120 mm): Thermally insulated boundary — zero heat flux condition ($q = 0$). This reflects the assumption that the outer chamber is well-insulated.
- IR Lamps (each): A heat generation rate of 40,000 W (40 kW) per lamp was applied as a volumetric heat source within each lamp domain. The lamps are modeled as diffuse surfaces with emissivity $\varepsilon = 0.99$.
- Silicon Wafer Surfaces: Surface-to-surface radiation boundary with emissivity $\varepsilon = 0.5$ applied on the wafer surfaces. The wafer interacts with the surrounding radiation environment.
- Quartz Tube: Built-in quartz (glass) material properties from COMSOL are used. Quartz is excluded from surface-to-surface radiation boundaries, as glass is a poor emitter of thermal radiation and tends to transmit and reflect incident radiation rather than emit it.
- Interior Boundaries: Continuity of heat flux across material interfaces is assumed (default in COMSOL).

3.3.3 Simulation Type

The simulation is conducted as a time-dependent (transient) analysis over a duration of 60 seconds. This captures the dynamic temperature evolution of the silicon wafer from its initial temperature (300 K) to its near-steady-state value under continuous IR lamp irradiation. A stationary study was first attempted to initialize the radiation field; the final time-dependent study uses Heat Transfer in Solids coupled with Surface-to-Surface Radiation.

3.4 Material Properties

Material properties for the simulation components are as follows:

Table 3: Material Properties Used in COMSOL Simulation

Material	Thermal Cond. [W/(m·K)]	Density [kg/m ³]	Heat Capacity [J/(kg·K)]	Emissivity
Silicon Wafer	Built-in (COMSOL)	Built-in (COMSOL)	Built-in (COMSOL)	0.5
IR Lamp	400	8700	10	0.99
Quartz Tube	Built-in (COMSOL)	Built-in (COMSOL)	Built-in (COMSOL)	Excluded

3.5 Simulation Results and Analysis

The COMSOL time-dependent simulation was successfully executed after meshing the geometry with a physics-controlled mesh (fine element size to resolve thermal gradients in the thin silicon wafer and near lamp surfaces). The simulation ran from $t = 0$ s to $t = 60$ s. The following results were obtained:

3.5.1 Temperature Distribution

The surface temperature distribution across the 2D cross-section at $t = 60$ s is shown in Figure 2. The color map ranges from approximately 400 K (dark purple, center) to approximately 1,900 K (bright yellow, near the IR lamp positions). The IR lamps, arranged in a circular pattern at 80 mm from the chamber center, are clearly visible as high-temperature zones. Heat is transferred to the silicon wafer primarily through surface-to-surface radiation across the quartz tube gap. The quartz tube acts as a transparent enclosure, allowing IR radiation to pass through with minimal absorption. A distinct radial temperature gradient is observed: temperatures are highest at the lamp surfaces and decay inward toward the wafer. At $t = 60$ s, the wafer region has reached an elevated temperature consistent with the transient point-probe data, confirming effective radiative coupling between the lamp array and the wafer.

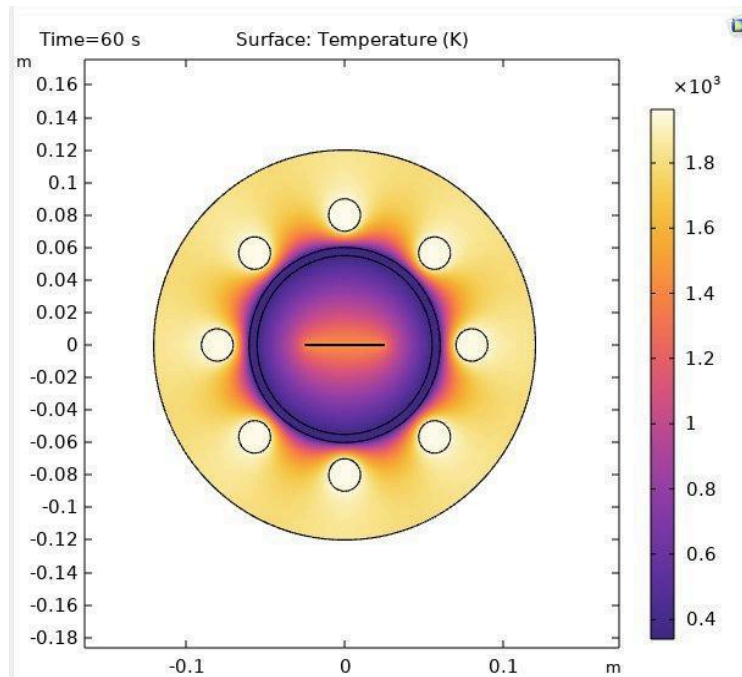


Figure 2: Surface temperature distribution in the RTP chamber at $t = 60$ s (time-dependent simulation). Temperature scale ranges from ~ 400 K (dark purple, center) to $\sim 1,900$ K (bright yellow, near lamp positions). The eight IR lamps are visible as high-temperature rings at $r = 80$ mm.

3.5.2 Wafer Midpoint Temperature

The temperature at a point on the silicon wafer (coordinate (0, 0) — the wafer midpoint) was recorded as a function of time using a Point Graph in COMSOL. The results, shown in Figure 3, demonstrate a rapid initial heating rate followed by an asymptotic approach to near-steady-state. Starting from the initial condition of 300 K, the wafer midpoint temperature rises steeply in the first 5–10 seconds, driven by the intense IR radiation from each 40,000 W lamp. By $t = 60$ s, the midpoint temperature reaches approximately 1,487 K ($\approx 1,214^\circ\text{C}$), well within the process window required for dopant activation and annealing. The x-axis range of the plot spans from -1.65 to 61.65 s and the y-axis from 257.7 K to 1,487.9 K, consistent with the COMSOL axis settings. The rapid temperature rise confirms the effectiveness of the eight-lamp circular array in delivering heat to the wafer through radiative coupling.

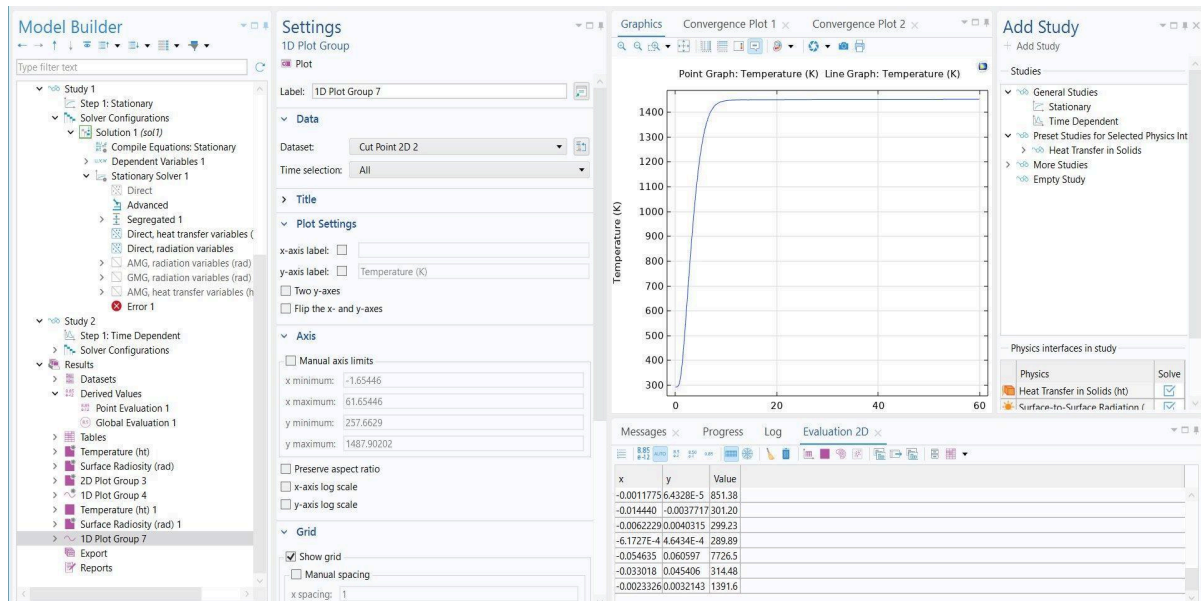


Figure 3: Wafer midpoint temperature vs. time ($t = 0\text{--}60$ s). Temperature rises rapidly from 300 K to approximately 1,487 K ($\sim 1,214^\circ\text{C}$) within 60 seconds, demonstrating the rapid heating capability of the IR lamp array. Dataset: Cut Point 2D 2; Physics: Heat Transfer in Solids (ht).

3.5.3 Heating Efficiency

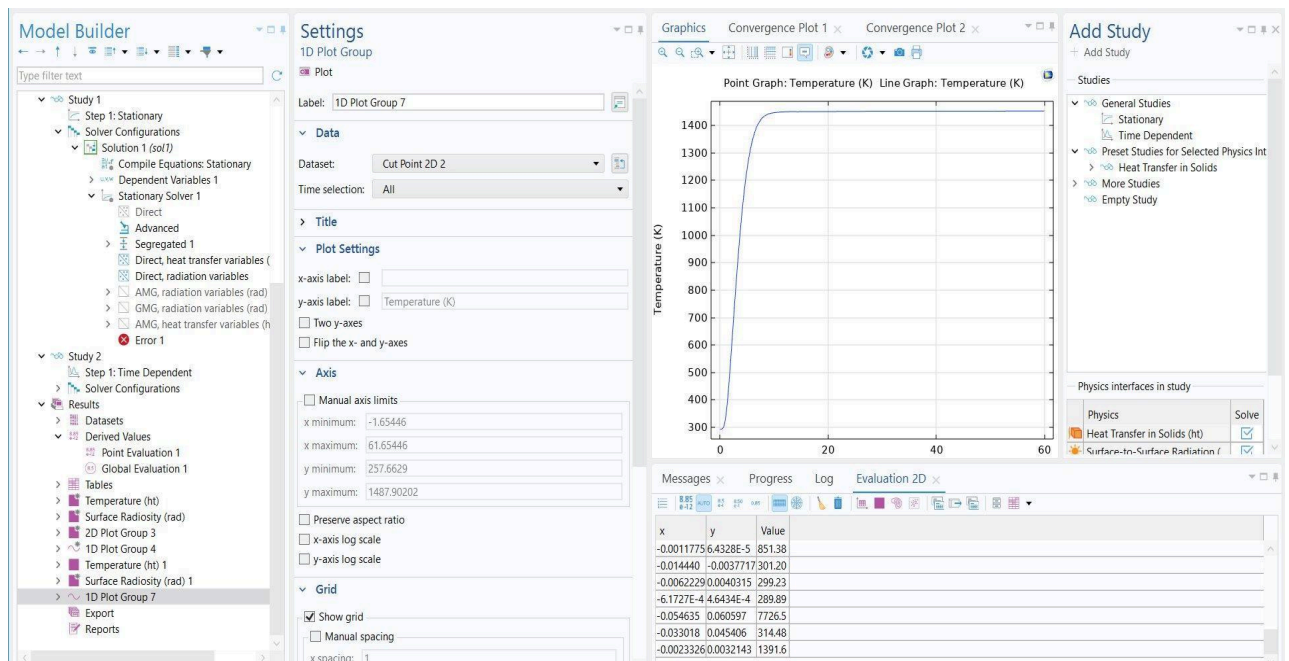


Figure 4: COMSOL Model Builder showing the two-study setup: Study 1 (Stationary, used for radiation field initialization) and Study 2 (Time Dependent, 0–60 s). The 1D Plot Group 7 displays the wafer midpoint temperature-time curve, with the Evaluation 2D table showing extracted point values across the domain at various coordinates.

The time-dependent simulation confirms that at elevated temperatures ($>1000^\circ\text{C}$), surface-to-surface radiation is the dominant heat transfer mechanism within the RTP

chamber. The high emissivity of the IR lamps ($\epsilon = 0.99$) maximizes radiative emission, while the wafer emissivity ($\epsilon = 0.5$) determines the fraction of incident radiation absorbed. The simulated heating rate is consistent with industry-standard RTP ramp rates (10–250°C/s): the wafer rises from 300 K to approximately 1,487 K over 60 seconds, implying an average ramp rate of ~19.8°C/s. The fastest heating occurs in the first 10 seconds (ramp rate >100°C/s), after which the rate slows as the wafer approaches thermal equilibrium with its radiative environment. These results validate the use of COMSOL's Heat Transfer in Solids with Surface-to-Surface Radiation physics for accurately modeling transient RTP thermal behavior.

4. Conclusion

This report has provided a comprehensive analysis of Rapid Thermal Processing (RTP), covering its critical importance in modern semiconductor manufacturing, fundamental physical mechanisms, engineering design approaches, system components and material selection rationale, and key operational challenges.

RTP has evolved into an essential process module in VLSI fabrication, enabling dopant activation, gate oxide growth, silicide formation, and other critical annealing steps with minimal thermal budget and maximum process precision. The challenges of temperature uniformity, pyrometric measurement accuracy, wafer slip, and system scalability continue to drive innovation in RTP hardware and process control.

The COMSOL Multiphysics time-dependent simulation of a simplified 2D RTP cross-section demonstrated the dominance of radiative heat transfer at high temperatures, consistent with the Stefan-Boltzmann law. The simulation successfully modeled the IR lamp heating mechanism, surface-to-surface radiation between diffuse surfaces, and the resulting transient temperature evolution in the silicon wafer. Starting from 300 K, the wafer midpoint temperature reaches approximately 1,487 K (~1,214°C) after 60 seconds, achieving an average ramp rate of ~19.8°C/s, in agreement with typical RTP operating conditions. These results provide valuable insights into RTP thermal behavior and validate the simulation methodology as a tool for RTP process design and optimization.

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